

AI-ASSISTED MULTILINGUAL COGNITIVE SCREENING & MEMORY ASSISTANCE PLATFORM

MASTER PROJECT CONTEXT • TEAM BUILD DOCUMENT

Purpose: give the whole team one shared understanding of what we are building, why, how the pipeline works, and what each module needs to do.

1. PROJECT IN ONE PARAGRAPH

We are building a web-based AI-assisted cognitive screening platform for older adults. The core research problem is to use speech produced during standardized cognitive/language tasks (especially picture description) to extract measurable acoustic, speech, linguistic and semantic indicators that can help distinguish cognitively impaired/dementia speech patterns from controls. The system is intended for screening and monitoring support, not as a standalone medical diagnosis. The final product combines the research model with a usable assessment interface, multilingual support, explainable results, longitudinal tracking, and selected cognitive/memory-assistance features inspired by SIH Problem Statement 26003.

2. WHAT MA'AM ACTUALLY GAVE US

- Ma'am has a classified/private dataset of voice recordings from dementia patients and control participants.
- Participants were shown a picture and asked to describe it / perform a related speech task.
- The dataset is difficult to obtain and is not currently in our hands, so the final model cannot honestly be trained/evaluated until we receive it.
- The recordings may contain multiple languages. The system therefore needs a multilingual strategy rather than assuming English.
- An important observation from ma'am: elderly speakers may have pronunciation deviations (for example, saying a word differently from the expected pronunciation).
- The intended end product is not merely a trained notebook model: it is a website that conducts an assessment, records responses, processes them, and presents an assessment/profile.

LABELS MATTER: Do not promise specific dementia stages unless the dataset actually contains reliable stage labels. If the dataset only contains dementia vs control labels, our model can only legitimately learn that distinction.

3. WHAT WE ARE TRYING TO SOLVE

The practical problem is that cognitive decline can affect how a person speaks, not only what they say. Relevant signals may include:

- Speech fluency and continuity
- Pause/silence behaviour and response latency
- Speech rate and duration
- Repetitions and hesitations
- Vocabulary/lexical diversity

- Sentence structure and linguistic complexity
- Semantic relevance to the task
- Whether expected objects/concepts are mentioned in a picture-description task
- Acoustic/prosodic properties such as pitch and energy
- Pronunciation/phonetic deviations where the available data supports measuring them

The central idea is therefore: do not feed raw audio into a black box and simply output 'dementia'. Instead, extract interpretable evidence from multiple layers of speech, combine those features, and use an ML model to estimate risk/class membership.

4. CORE SYSTEM PIPELINE

High-level flow:

USER → WEB ASSESSMENT → STANDARDIZED TASK → AUDIO RECORDING → AUDIO PREPROCESSING → SPEECH/ACOUSTIC FEATURES + ASR → NLP/SEMANTIC FEATURES → FEATURE FUSION → ML MODEL → RISK/PROFILE → DASHBOARD

Stage	Input	What happens	Output
1. Assessment UI	Prompt/image	Shows task and instructions in selected language	User response
2. Recording	Microphone	Browser records speech	Audio file
3. Audio processing	Audio	Clean/standardize audio and calculate basic properties	Processed audio + acoustic features
4. Speech recognition	Audio	Convert speech to text using suitable ASR	Transcript
5. Speech/NLP analysis	Audio + transcript	Measure fluency, pauses, vocabulary, semantics, repetitions, etc.	Feature vector
6. ML inference	Feature vector	Apply trained classifier	Class/risk estimate
7. Explanation	Model + features	Show which measurable factors contributed	Cognitive/speech profile
8. Storage	Assessment result	Save history securely	Longitudinal record

5. THE THREE ANALYSIS LAYERS

5.1 Acoustic / Paralinguistic Layer

This asks: HOW did the person speak?

- Audio duration
- Speech/silence ratio
- Pause count and pause duration
- Response latency before starting
- Speech rate / words per unit time where transcript alignment permits
- Pitch/prosody
- Energy/intensity
- MFCCs and related spectral features

- Other robust acoustic descriptors if justified by experiments

Mental model: audio → numbers. MFCCs are one way of converting properties of the audio spectrum into numerical features usable by ML.

5.2 ASR / Transcript Layer

This asks: WHAT did the person say?

- Speech-to-text transcript
- Word count
- Sentence count
- Repetitions
- Filler/hesitation markers when ASR preserves them
- Vocabulary usage
- Task-specific object/concept mentions
- Pronunciation errors only where the ASR/phonetic pipeline can measure them reliably

5.3 NLP / Semantic Layer

This asks: WHAT does the response mean and how well does it match the task?

- Semantic similarity between response and expected task content
- Coverage of relevant objects/concepts in a picture
- Lexical diversity
- Repetition / redundancy
- Sentence/utterance complexity
- Coherence/relevance
- Language-specific or multilingual embedding features

MULTILINGUAL: Feature definitions must be validated against the actual dataset and language coverage. We should not assume an English-only NLP metric works equally well in every language.

6. FEATURE FUSION + MODEL

The strongest practical baseline is to combine the engineered features from the above layers into one feature vector, then train a conventional supervised classifier. This is preferable to immediately training a huge deep network from scratch, especially if the private dataset is not enormous.

- Candidate baselines: Logistic Regression, SVM, Random Forest, Gradient Boosting/XGBoost (depending on dataset size and validation).
- Compare models rather than choosing one arbitrarily.
- Use feature scaling where required.
- Use cross-validation during development.
- Tune hyperparameters only after establishing a clean baseline.
- Later, if the dataset supports it, compare against pretrained speech representations / transformer-based approaches.

DO NOT OVERENGINEER: The final algorithm is a research result, not a decision we should lock before seeing the dataset.

7. DATA SPLITTING — CRITICAL

If a participant has multiple recordings, all recordings from that participant must stay in the same train/validation/test partition. Otherwise the model can accidentally learn the speaker's identity/voice characteristics instead of cognitive status. This is called speaker leakage and can make results look artificially excellent.

- Prefer participant-level splitting.
- Keep a true held-out test set.
- Do not repeatedly tune against the final test set.
- Document every preprocessing and split decision.

8. EVALUATION

Accuracy alone is not enough for a medical screening-style classifier. Report:

- Sensitivity / Recall — how many relevant impaired cases are detected
- Specificity — how many controls are correctly identified
- Precision
- F1 score
- ROC-AUC where appropriate
- Confusion matrix
- Class distribution
- Cross-validation results

We should also inspect performance by language, task type, age/other available demographic groups, and recording quality where the dataset contains enough samples to do so.

9. MULTILINGUAL DESIGN

The multilingual requirement affects both the AI pipeline and the UI.

- The user selects a preferred language or the system performs language identification where reliable.
- Assessment prompts/images are presented with localized instructions.
- Use a multilingual ASR model/service appropriate to the languages actually present in the dataset.
- Use multilingual NLP/embedding models or language-specific processing where required.
- Do not translate every transcript to English automatically and assume meaning/linguistic features are preserved.
- Maintain language metadata with every recording and prediction.
- Evaluate model performance separately by language before claiming multilingual performance.

10. FRONTEND / WEBSITE — WHAT THE USER SEES

Recommended initial approach: simple HTML/CSS/JavaScript frontend connected to a Python API. React can be introduced if the team already needs it, but it is not required to prove the core system.

10.1 Landing / Home

- Clear project purpose
- Simple explanation: AI-assisted cognitive screening and memory support
- Start assessment button

- Language selector
- Accessibility-first design: large text, high contrast, simple navigation, minimal clutter

10.2 User / Patient Setup

- Participant ID / basic non-sensitive metadata as permitted by ma'am
- Language selection
- Microphone permission check
- Optional webcam permission only if the feature is actually implemented and ethically/technically justified
- Audio test

10.3 Assessment Screen

- One task at a time
- Large image/prompt
- Clear voice/text instruction
- Record / stop controls
- Recording timer
- Retake policy if allowed by the protocol
- Progress indicator
- Do not make the interface feel like a stressful exam

10.4 Suggested Assessment Tasks

1. Picture description — primary task because it aligns with the available dataset.
2. Object naming — identify shown objects.
3. Immediate recall — remember a small set of items.
4. Delayed recall — reproduce them later.
5. Category/verbal fluency — name as many items as possible from a category in a fixed time.
6. Optional cognitive games — memory, attention, pattern/object recognition, etc., if validated and useful.

SCOPE: The exact task set should follow the research protocol and what ma'am approves. We should not invent clinical validity for our own games.

10.5 Results / Clinician Dashboard

- Overall screening risk / model output
- Confidence/probability with careful wording
- Speech metrics
- Linguistic/semantic metrics
- Task-by-task performance
- Timeline/history of previous assessments
- Language and task metadata
- Model explanation / feature contribution
- Flag for clinician review rather than a definitive diagnosis

11. WEBCAM / FACIAL EXPRESSIONS — HOW TO HANDLE THIS

The final website may include webcam capability because the desired platform can observe additional behavioural information, but our current classified dataset is voice-based. Therefore we cannot claim that facial expressions are part of the dementia classifier unless we obtain appropriately labeled video data.

- Webcam can initially be treated as an optional interface/behavioural observation module.

- Do not train or advertise a facial-dementia model without labeled video data.
- If future labeled video data becomes available, a separate computer-vision branch can be researched and validated.
- Always consider consent, privacy, storage and ethics before recording video.

12. MEMORY ASSISTANCE / SIH 26003 FEATURES

Ma'am has said that useful parts of SIH Problem Statement 26003 can be incorporated. These features should complement the core speech-screening research rather than distract from it.

- Cognitive games: memory, attention, routine recall, pattern/object recognition, emotional engagement.
- Adaptive difficulty: increase/decrease task difficulty based on performance.
- Multilingual voice assistance.
- Culturally familiar themes/content where appropriate.
- Reminders: medication, hydration, activities and appointments.
- Caregiver/health-worker dashboard.
- Longitudinal progress tracking.
- Mobile/tablet-friendly interface.
- Offline or low-connectivity support if feasible.
- Secure data handling.

PRIORITY: Prioritize the core assessment + model first. Add SIH features as modular extensions so the project remains deliverable.

13. PROPOSED TECH STACK

Layer	Recommended	Reason
Frontend	HTML + CSS + JavaScript initially	Enough for recording, task flow, API calls and dashboard
Optional frontend upgrade	React	Useful if UI becomes complex/component-heavy
Backend	Python + FastAPI	Natural bridge to audio/ML ecosystem
ML	Python, NumPy, Pandas, scikit-learn	Feature engineering + baseline classifiers
Audio	librosa + standard audio utilities	Acoustic/spectral feature extraction
ASR	Suitable pretrained multilingual ASR	Speech → transcript
NLP	Transformers / multilingual embeddings as appropriate	Semantic and linguistic analysis
Database	SQLite initially; PostgreSQL later if needed	Assessment/user/history storage
Charts	Chart.js or equivalent	Clinician dashboard visualization
Version control	Git + GitHub	Team collaboration and history

14. API / BACKEND CONCEPT

The website and AI system should communicate through a clean API boundary:

- Frontend sends selected task/language + recorded audio to backend.

- Backend validates requests and stores/processes the audio according to the project protocol.
- Audio pipeline extracts features and/or generates transcripts.
- ML service produces prediction and feature-level explanation.
- Backend returns structured JSON.
- Frontend renders results and stores an assessment record.

Example conceptual response:

```
{"assessment_id": "...", "language": "...", "risk_score": 0.72, "speech_features": {...},  
"linguistic_features": {...}, "model_version": "...", "needs_clinician_review": true}
```

15. DATA MODEL / STORAGE IDEA

- Participant: participant_id, permitted metadata
- Assessment: assessment_id, participant_id, timestamp, language, protocol/version
- Task response: task_id, audio reference, transcript, task score
- Extracted features: acoustic + linguistic + semantic features
- Prediction: model version, class/risk output, confidence/probability, explanation
- History: previous assessments for longitudinal tracking
- Consent/privacy metadata where required

PRIVACY: Avoid storing raw audio/video longer than necessary. Access controls and anonymization/pseudonymization should be considered from the beginning.

17. TEAM MODULE BREAKDOWN

Workstream	Responsibilities	Deliverables
Data + ML	Dataset audit, preprocessing, feature extraction, splitting, baselines, evaluation	Notebooks/scripts, trained model, metrics
Speech + NLP	ASR, transcript cleaning, linguistic/semantic features, multilingual experiments	Speech/NLP pipeline + feature extractor
Backend	FastAPI, upload/inference endpoints, database integration	API + model-serving layer
Frontend	Assessment UI, recorder, task flow, dashboard	Responsive web interface
SIH / Product	Games, reminders, adaptive logic, caregiver workflow	Modular support features
Testing + Documentation	Integration testing, edge cases, reproducibility, report/demo	Test plan, architecture docs, presentation

18. WHAT WE CAN BUILD BEFORE GETTING MA'AM'S DATA

- Complete frontend assessment flow using mock tasks.
- Implement microphone recording with browser APIs.
- Build backend file-upload endpoint.
- Create audio preprocessing and feature extraction pipeline.
- Create ASR → transcript pipeline with a temporary/public dataset for technical testing if needed.
- Build NLP/semantic feature extraction.

- Create a mock ML inference endpoint returning structured results.
- Build clinician dashboard and longitudinal-history UI using mock JSON.
- Set up Git/GitHub branching and project structure.

DATA HONESTY: Temporary/public/synthetic data can validate the software pipeline, but it must never be presented as evidence for the final dementia model.

19. PROJECT FOLDER STRUCTURE

A practical initial repository could look like this:

```
project/
├── frontend/
│   ├── index.html
│   ├── styles.css
│   └── app.js
├── backend/
│   ├── main.py
│   ├── routes/
│   ├── services/
│   └── schemas/
├── ml/
│   ├── preprocessing/
│   ├── features/
│   ├── asr/
│   ├── nlp/
│   ├── training/
│   └── inference/
├── data/
│   └── README.md # no private dataset committed
├── models/
├── tests/
├── docs/
└── README.md
```

21. THE MINIMUM VIABLE PRODUCT (MVP)

7. The user selects the language.
8. The website displays a standardized picture/task.
9. The browser records the participant's response.
10. Audio is uploaded to the backend.
11. Backend runs preprocessing + feature extraction.
12. ASR generates transcripts where supported.
13. The model combines validated features and generates a screening estimate.
14. Dashboard displays task results, key speech/linguistic indicators and model output.
15. Assessment is stored for longitudinal comparison.

Everything else is an extension.

22. THE IDEAL FINAL DEMO

A participant enters the platform → selects language → completes a short guided assessment → the browser records each response → backend processes the recordings → AI extracts acoustic + linguistic + semantic evidence → model generates a screening risk/profile → clinician dashboard shows interpretable results and history → the platform optionally recommends approved cognitive exercises/reminders based on the user's profile.

23. RESEARCH QUESTIONS WE NEED TO ANSWER

- Which acoustic features differ most between the available classes?
- Does transcript/NLP information improve performance over acoustic features alone?
- Does combining acoustic + linguistic + semantic features improve generalization?
- How does performance vary by language?
- Which tasks are most informative?
- Which model gives the best balance of performance and interpretability?
- Are predictions robust to recording quality and speaker variability?
- Can the system produce a useful longitudinal profile rather than only one binary result?

24. IMMEDIATE TEAM CHECKLIST

- ☐ Get the dataset requirements/labels from ma'am.
- ☐ Confirm exactly which languages are present.
- ☐ Confirm whether each participant has one or multiple recordings.
- ☐ Confirm the exact task/prompt used for every recording.
- ☐ Determine class distribution and available demographic metadata.
- ☐ Determine allowed use/storage of recordings.
- ☐ Set up GitHub repository and branch ownership.
- ☐ Build browser recording prototype.
- ☐ Build Python audio-feature extraction prototype.
- ☐ Build an ASR prototype for the actual languages.
- ☐ Create participant-level data split.
- ☐ Train the first interpretable baseline.
- ☐ Build a mock dashboard while ML work progresses.

26. IMPORTANT LIMITATIONS

- The private dataset is the foundation of the final research result; without inspecting it, exact labels, languages, model architecture and performance cannot be finalized.
- Speech-based cognitive screening can be affected by age, accent, education, hearing, microphone quality, language, fatigue and other factors.
- A model's statistical performance does not automatically establish clinical validity.
- Any real deployment involving patient data would require appropriate ethics, privacy, consent, security and clinical validation.
- The project should be presented as an AI-assisted screening/research prototype unless and until proper clinical validation exists.

27. SOURCE / DESIGN BASIS

This team document consolidates the project requirements discussed with ma'am, the team's interpretation of the private voice-dataset task, and the previously discussed SIH 26003 feature ideas. The web-development and ML roadmaps are being used only as supporting engineering guidance: the web roadmap emphasizes frontend/backend separation, APIs, Git, JavaScript, databases and deployment; the ML roadmap emphasizes data handling, supervised learning, evaluation metrics, cross-validation, model selection and deployment.

Engineering note from the roadmaps: web applications benefit from a clear frontend/backend boundary and API-based communication, while ML development should include train/test separation, cross-validation, model comparison and metrics such as precision, recall, F1 and ROC-AUC.